Interior and Exterior Angles of a Triangle

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	6 — Relationships in Triangles
Lesson	Thread B · Day 2 — Interior and Exterior Angles of a Triangle	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.GR.1.5

2. Standards

MA.8.GR.1.5 Solve problems involving the relationships of interior and exterior angles of a triangle.

Must. Find a missing interior angle with the Triangle Sum Theorem; find an exterior angle, or a remote interior angle, with the Exterior Angle Theorem; write and solve the equation when the angles are expressions.

Must not. Angle sums of polygons (MA.8.GR.1.6); parallel lines and transversals; reflex angles (the angle around the outside is a named error, not an answer).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes I–II: 180° as torn-off corners, as a sum and as a linear pair
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 7 split; the re-vote
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes II: the theorem from the triangle sum and the linear pair together

4. Learning Target and Essential Question

Learning target. I can find a missing angle of a triangle, inside or outside, with the Triangle Sum and the Exterior Angle Theorem.

Essential question. Why do the three angles of every triangle add to the same number?

Building on. T-B1 — supplementary angles and the linear pair; Grade 4 (MA.4.GR.1.3) — angle measures add; Unit 1 — equations in one unknown.

Working toward. Unit 11 — equations and systems; Geometry — the angle sums of polygons and triangle proofs.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	10	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	12	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (12 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Can a triangle have two right angles? Two obtuse angles? Why not?	DOK 2
Why does an exterior angle equal the two remote interior angles? Use the linear pair.	DOK 3
Each corner of a triangle has two exterior angles. Why are they equal?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	Two angles of a triangle are 47° and 68°. Find the third angle.	***65°***	115 — stopped at the sum [Math Nation names misremembering the sum]; 245 — used 360 [the Triangle Sum is 180]
2	An isosceles triangle has two base angles of 52°. Find the third angle.	***76°***	128 — used one 52 [the two base angles are equal: both count]; 52 — took all three as equal [isosceles: two equal, not three]
3	A triangle's remote interior angles are 38° and 77°. Find the exterior angle x.	***115°***	65 — the interior angle beside it, 180 − 115 [Source of Truth, GR.1.5: the exterior angle confused with its neighbour]; 245 — the angle around the outside [Source of Truth, GR.1.5: the reflex angle]
4	A triangle's angles are 45°, 60° and 75°. What is the exterior angle beside the 75°?	***B — 105°***	(A) equals the adjacent interior angle [the brief's named error: the exterior angle is not the interior angle beside it]; (C) the angle around the outside, 360 − 75 [Source of Truth, GR.1.5: the exterior angle taken as the reflex angle]; (D) only one remote angle [the Exterior Angle Theorem adds both]
5	An exterior angle of a triangle is 130°. One remote interior angle is 55°. Find the other remote interior angle.	***75°***	185 — added the exterior and the remote angle [the theorem: the remote angles add TO the exterior one]; 50 — the interior angle beside the exterior one [Source of Truth, GR.1.5]
6	A triangle's angles are x°, $3x^\circ$ and $(2x+12)^\circ$. Find each angle.	***28°, 84° and 68°***	
7	An exterior angle measures (5x + 4)°; its remote interior angles measure 52° and (3x)°. Which equation finds x?	***C — 5x + 4 = 52 + 3x***	(A) the exterior angle used as an interior angle in the Triangle Sum [Math Nation names writing the equation for the wrong relationship]; (B) one remote angle only [the theorem adds both]; (D) all four angles summed to 180 [the exterior angle is outside the triangle's sum]
8	A triangle has angles of 41° and 66°, and its longest side is 9 cm. Find its third angle.	***73°***	64 — subtracted the 9 as well [the extra number]; 107 — stopped at the sum
9	Why is an exterior angle of a triangle always bigger than either of its remote interior angles? Explain.	***It is the two remote angles added, each more than 0°.***	'Because it is on the outside' [no reason]; 'it is not always bigger' [the remote angles are each more than 0°]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. Two lines on the desk: inside, the three make 180; outside, the two far ones added. Shade the two remote angles before writing the equation.

ELL. **Interior**: inside. **Exterior**: outside. **Remote**: far away — not touching. Point to each with the word.

Enrichment. A triangle's angles are in the ratio 2 : 3 : 4. Find each angle, then the exterior angle at the largest one. (40°, 60°, 80°; 100°.)

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: an exterior angle is the two remote interior angles added, and each of them is more than 0°, so the sum is bigger than either one. A board that says only 'because it is outside' has the claim and not the reason.

Vocabulary. **interior angle**; **exterior angle**; **remote interior angles**; **Triangle Sum Theorem**

Materials. Whiteboards and markers; one paper triangle to tear for Notes I, if there is time. Calculators allowed, printed nowhere.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Find missing angles in triangles — JFJ; Exterior Angle Theorem — FMP; Triangle Angle-Sum Theorem — 6Q6.